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Contents

Defined, *design* is "the arrangement of elements or details in a product or work of art." Of course, given a large number of elements, they can be arranged in an almost endless way with a plethora of results. Many things can impact the design of your real-time character, but the most impacting and unavoidable one is *limitations*.

Limitations

By limitations I don't mean limitations on your imagination. I mean hard limitations like making a convincing human or creature out of a thousand triangles or less. However, with the growth of the hardware and computing power of the PC and the next generation of consoles like Xbox 2 and PlayStation 3, polygon limitations are becoming less severe. Still, characters for real-time games can rarely afford to have flowing gowns or octopoid tentacles. Organic shapes that undulate and look "squishy" are hard to pull off, as is cloth or long, flowing strands of hair when you have to represent them with triangles. Designing real-time characters requires much more thought than a character destined *only* to be rendered, as in a movie. With lower-polygon meshes it's usually best to keep the designs compact and simple using transparency mapped onto two-sided, segmented planes to create the illusion of hair or scraps of cloth.

Even as polygon limitations ease and characters that are 10,000 triangles or more are possible, other limitations like number of bones in a skeleton, number of possible morph targets, and amount of texture resolution will always exist. For example, this real-time character done for a recent XNA demo for the Xbox is over 80,000 triangles: